

Operate Cloud Storage and Diagnose Cloud SQL

105 minutes · Mapped criteria A4, K5

Purpose and work product

Apply storage lifecycle controls, load mock customer data and diagnose a simulated Cloud SQL connection bottleneck.

Submit: lifecycle policy, query evidence and database incident report

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
customers.csv	Mock data, script or configuration used by this activity
lifecycle.json	Mock data, script or configuration used by this activity
diagnose.sql	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Set BUCKET to a globally unique lower-case name and create a regional bucket with uniform bucket-level access.

Step 2

Upload customers.csv and verify object size, generation and checksum.

Step 3

Apply lifecycle.json only after reviewing the 30-day condition and Nearline action.

Step 4

Enable versioning and upload a corrected copy of one record; list object generations.

Step 5

For the database path, use the trainer-provided Cloud SQL sandbox or mock outputs if live provisioning is not approved.

Step 6

Load the sample customer records into a training table and run row-count and region-grouping validation queries.

Step 7

Run diagnose.sql or interpret the supplied mock results to inspect connection states and slow statements.

Step 8

Reproduce a safe connection-pool saturation symptom with the trainer-provided harness.

Step 9

Apply one bounded remediation such as pool sizing, query index or instance connection limit, then repeat the same test.

Step 10

Document before/after connection count, mean execution time and error rate.

Step 11

Export the final query evidence and database incident report.

Step 12

Delete database resources if individually provisioned; retain only approved shared sandbox assets.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.